

Exposure-Efficacy Analysis of Antibody-Drug Conjugates Delivering an Excessive Level of Payload to Tissues.

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Antibody-drug conjugates (ADCs) contain a disease-receptor antibody and a payload drug connected via a linker. The payload delivery depends on both tumor properties and ADC characteristics. In this study, we used different linkers, attachment sites, and doses to modulate payload delivery of several ADCs bearing maytansinoids (e.g., DM1), auristatins (e.g., MMAE), and DNA alkylating agents [e.g., pyrrolo[2,1-c][1,4]benzodiazepine-dimer (PBD)] as payloads in HER2- or CD22-expressing xenograft models. The tumor growth inhibition and ADC stability and exposure data were collected and analyzed from these dosed animals. The trend analysis suggests that intratumoral payload exposures that directly related the combination of conjugate linker and dose correlate with the corresponding efficacies of three payload types in two antigen-expressing xenograft models. These preliminary correlations also suggest that a minimal threshold concentration of intratumoral payload is required to support sustained efficacy. In addition, an ADC can deliver an excessive level of payload to tumors that does not enhance efficacy ("Plateau" effect). In contrast to tumor payload concentrations, the assessments of systemic exposures of total antibody (Tab) as well as the linker, dose, site of attachment, plasma stability, and drug-to-antibody ratio changes of these ADCs did not consistently rationalize the observed ADC efficacies. The requirement of a threshold payload concentration for efficacy is further supported by dose fractionation studies with DM1-, MMAE-, and PBD-containing ADCs, which demonstrated that single-dose regimens showed better efficacies than fractionated dosing. Overall, this study demonstrates that 1) the linker and dose together determine the tissue payload concentration that correlates with the antitumor efficacy of ADCs and 2) an ADC can deliver an unnecessary level of payload to tumors in xenograft models.

Antibody-drug conjugates: Recent advances in payloads.

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Antibody–drug conjugates (ADCs), which combine the advantages of monoclonal antibodies with precise targeting and payloads with efficient killing, show great clinical therapeutic value. The ADCs' payloads play a key role in determining the efficacy of ADC drugs and thus have attracted great attention in the field. An ideal ADC payload should possess sufficient toxicity, low immunogenicity, high stability, and modifiable functional groups. Common ADC payloads include tubulin inhibitors and DNA damaging agents, with tubulin inhibitors accounting for more than half of the ADC drugs in clinical development. However, due to clinical limitations of traditional ADC payloads, such as inadequate efficacy and the development of acquired drug resistance, novel highly efficient payloads with diverse targets and reduced side effects are being developed. This perspective summarizes the recent research advances of traditional and novel ADC payloads with main focuses on the structure-activity relationship studies, co-crystal structures, and designing strategies, and further discusses the future research directions of ADC payloads. This review also aims to provide valuable references and future directions for the development of novel ADC payloads that will have high efficacy, low toxicity, adequate stability, and abilities to overcome drug resistance.

Delivery of Monomethyl Auristatin F to the Tumor Microenvironment with

Noninternalizing Fibroblast Activation Protein-Cleavable Small Molecule-Drug Conjugates Elicits Potent In Vivo Anticancer Activity.

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OncoFAP is an ultrahigh affinity ligand of fibroblast activation protein (FAP), a tumor-associated antigen overexpressed in the stroma of the majority of solid tumors. OncoFAP has been previously implemented as a tumor-homing moiety for the development of small molecule drug conjugates (SMDCs). In the same context, the glycine--proline dipeptide was included with the aim to selectively undergo cleavage only in the presence of the target FAP, triggering the consequent release of the cytotoxic payload in the tumor microenvironment. In this work, we evaluate the use of monomethyl auristatin F (MMAF) as a payload, a close derivative of MMAE bearing a charged carboxylic acid that hampers its cellular permeability, typically employed in the development of internalizing antibody-drug conjugates. The novel OncoFAP-GlyPro-MMAF and the previously described OncoFAP-GlyPro-MMAE were compared in a head-to-head therapeutic experiment in mice bearing FAP-positive tumors. Surprisingly, the MMAF conjugate mediated potent antitumor activity, despite its poor cellular permeability.

Peptide-drug conjugate-based novel molecular drug delivery system in cancer.

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Drug delivery systems are generally believed to comprise drugs and excipients. A peptide-drug conjugate is a single molecule that can simultaneously play multiple roles in a drug delivery system, such as in vivo drug distribution, targeted release, and bioactivity functions. This molecule can be regarded as an integrated drug delivery system, so it is called a molecular drug delivery system. In the context of cancer therapy, a peptide-drug conjugate comprises a tumor-targeting peptide, a payload, and a linker. Tumor-targeting peptides specifically identify membrane receptors on tumor cells, improve drug-targeted therapeutic effects, and reduce toxic and side effects. Payloads with bioactive functions connect to tumor-targeting peptides through linkers. In this review, we explored ongoing clinical work on peptide-drug conjugates targeting various receptors. We discuss the binding mechanisms of tumor-targeting peptides and related receptors, as well as the limiting factors for peptide-drug conjugate-based molecular drug delivery systems.

Benzyl Ammonium Carbamates Undergo Two-Step Linker Cleavage and Improve the Properties of Antibody Conjugates.

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Targeted payload delivery strategies, such as antibody-drug conjugates (ADCs), have emerged as important therapeutics. Although considerable efforts have been made in the areas of antibody engineering and labeling methodology, improving the overall physicochemical properties of the linker/payload combination remains an important challenge. Here we report an approach to create an intrinsically hydrophilic linker domain. We find that benzyl α -ammonium carbamates (BACs) undergo tandem 1,6-1,2-elimination to release secondary amines. Using a fluorogenic hemicyanine as a model payload component, we show that a zwitterionic BAC linker improves labeling efficiency and reduces antibody aggregation when compared to a commonly used para-amino benzyl (PAB) linker as well as a

cationic BAC. Cellular and in vivo fluorescence imaging studies demonstrate that the model payload is specifically released in antigen-expressing cells and tumors. The therapeutic potential of the BAC linker strategy was assessed using an MMAE payload, a potent microtubule-disrupting agent frequently used for ADC applications. The BAC-MMAE combination enhances labeling efficiency and cellular toxicity when compared to the routinely used PAB-Val-Cit ADC analogue. Broadly, this strategy provides a general approach to mask payload hydrophobicity and improve the properties of targeted agents.

Antibody-Drug Conjugates in Solid Tumor Oncology: An Effectiveness Payday with a Targeted Payload.

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Antibody-drug conjugates (ADCs) are at the forefront of the drug development revolution occurring in oncology. Formed from three main components—an antibody, a linker molecule, and a cytotoxic agent ("payload"), ADCs have the unique ability to deliver cytotoxic agents to cells expressing a specific antigen, a great leap forward from traditional chemotherapeutic approaches that cause widespread effects without specificity. A variety of payloads can be used, including most frequently microtubular inhibitors (auristatins and maytansinoids), as well as topoisomerase inhibitors and alkylating agents. Finally, linkers play a critical role in the ADCs' effect, as cleavable moieties that serve as linkers impact site-specific activation as well as bystander killing effects, an upshot that is especially important in solid tumors that often express a variety of antigens. While ADCs were initially used in hematologic malignancies, their utility has been demonstrated in multiple solid tumor malignancies, including breast, gastrointestinal, lung, cervical, ovarian, and urothelial cancers. Currently, six ADCs are FDA-approved for the treatment of solid tumors: ado-trastuzumab emtansine and trastuzumab deruxtecan, both anti-HER2; enfortumab-vedotin, targeting nectin-4; sacituzumab govitecan, targeting Trop2; tisotumab vedotin, targeting tissue factor; and mirvetuximab soravtansine, targeting folate receptor- α . Although they demonstrate utility and tolerable safety profiles, ADCs may become ineffective as tumor cells undergo evolution to avoid expressing the specific antigen being targeted. Furthermore, the current cost of ADCs can be limiting their reach. Here, we review the structure and functions of ADCs, as well as ongoing clinical investigations into novel ADCs and their potential as treatments of solid malignancies.

Blurring the Role of Oligonucleotides: Spherical Nucleic Acids as a Drug Delivery Vehicle.

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Nucleic acids are generally regarded as the payload in gene therapy, often requiring a carrier for intracellular delivery. With the recent discovery that spherical nucleic acids enter cells rapidly, we demonstrate that nucleic acids also have the potential to act as a delivery vehicle. Herein, we report an amphiphilic DNA-paclitaxel conjugate, which forms stable micellar nanoparticles in solution. The nucleic acid component acts as both a therapeutic payload for intracellular gene regulation and the delivery vehicle for the drug component. A bioreductively activated, self-immolative disulfide linker is used to tether the drug, allowing free drug to be released upon cell uptake. We found that the DNA-paclitaxel nanostructures enter cells ~100 times faster than free DNA, exhibit increased stability against nuclease, and show nearly identical cytotoxicity as free drug. These nanostructures allow one to access a gene

target and a drug target using only the payloads themselves, bypassing the need for a cocarrier system.

[Antibody-drug conjugates as new therapeutic agents in uro-oncology].

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Antibody-drug conjugates represent a new class of therapeutic agents that are already being used in the field of uro-oncology. They consist of an antibody directed against a specific tumour antigen linked to a cytotoxic substance ("payload") which acts after internalisation into the tumour cell and its release. Currently, approval in the European Union is restricted to enfortumab vedotin which is directed against nectin-4 and carries the microtubule-inhibiting active ingredient monomethyl auristatin E (MMAE). Enfortumab vedotin is approved for locally advanced or metastatic urothelial carcinoma in the third line of therapy after platinum-based chemotherapy and after therapy with a programmed cell death (ligand) 1 (PD-[L]1) immune checkpoint inhibitor. However, an expansion of the indication of enfortumab vedotin as monotherapy and in combination with PD-(L)1 immune checkpoint inhibitors, as well as approval of other antibody-drug conjugates is expected in the future. This could sustainably change the therapy sequence in urothelial carcinoma. Currently, several clinical trials are recruiting in different therapeutic settings. This article presents the new substance class of antibody-drug conjugates, their mechanism of action, their representatives and clinical studies, and points out practice-relevant side effects and how to deal with them.

Updates in the Development of ImmunoRNases for the Selective Killing of Tumor Cells.

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Targeted cancer therapy includes, amongst others, antibody-based delivery of toxic payloads to selectively eliminate tumor cells. This payload can be either a synthetic small molecule drug composing an antibody-drug conjugate (ADC) or a cytotoxic protein composing an immunotoxin (IT). Non-human cytotoxic proteins, while potent, have limited clinical efficacy due to their immunogenicity and potential off-target toxicity. Humanization of the cytotoxic payload is essential and requires harnessing of potent apoptosis-inducing human proteins with conditional activity, which rely on targeted delivery to contact their substrate. Ribonucleases are attractive candidates, due to their ability to induce apoptosis by abrogating protein biosynthesis via tRNA degradation. In fact, several RNases of the pancreatic RNase A superfamily have shown potential as anti-cancer agents. Coupling of a human RNase to a humanized antibody or antibody derivative putatively eliminates the immunogenicity of an IT (now known as a human cytolytic fusion protein, hCFP). However, RNases are tightly regulated in vivo by endogenous inhibitors, controlling the ribonucleolytic balance subject to the cell's metabolic requirements. Endogenous inhibition limits the efficacy with which RNase-based hCFPs induce apoptosis. However, abrogating the natural interaction with the natural inhibitors by mutation has been shown to significantly enhance RNase activity, paving the way toward achieving cytolytic potency comparable to that of bacterial immunotoxins. Here, we review the immunoRNases that have undergone preclinical studies as anti-cancer therapeutic agents.

Antibody-drug conjugate payloads.

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Toxin payloads, or drugs, are the crucial components of therapeutic antibody-drug conjugates (ADCs). This review will give an introduction on the requirements that make a toxic compound suitable to be used in an antitumoral ADC and will summarize the structural and mechanistic features of four drug families that yielded promising results in preclinical and clinical studies.
